ELSEVIER

# EXECUTIVE FORUM

# MEASURING PROGRESS IN ENTREPRENEURSHIP EDUCATION

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## EXECUTIVE SUMMARY

*This article presents the results of a survey that ranked university entrepreneurship programs. The survey also explored how universities determined what courses constituted a program in entrepreneurship and how they determined the criteria that impact an entrepreneurship program's quality. We conclude the article with a discussion of the education pilot criteria for the Malcolm Baldrige National Quality Award that may be useful for measuring progress in entrepreneurship education.*

*A mail survey was undertaken in late 1994. This survey was sent to deans at 941 business schools in the United States, 42 in Canada, and 270 overseas. Of the 311 replies, 233 came from U.S. business schools, 16 from Canadian schools, and 62 from schools in other countries.*

*The top seven criteria suggested for ranking entrepreneurship programs were courses offered, faculty publications, impact on community, alumni exploits, innovations, alumni start-ups, and outreach to scholars. The most frequently offered entrepreneurship courses at both the undergraduate and graduate levels in the entrepreneurship programs surveyed were entrepreneurship or starting new firms, small business management, field projects/venture consulting, starting and running a firm, venture plan writing, and venture finance.*

*The survey uncovered a number of problems with how academics ranked other entrepreneurship programs. Evaluators did not specify the criteria they used to rank entrepreneurship programs. Evalua-*

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Copies of the Malcolm Baldrige National Quality Award 1995 Education Pilot Criteria may be obtained from: Malcolm Baldrige National Quality Award, 1995 Educational Pilot program, National Institute of Standards and Technology, Route 270 and Quince Orchard Road, Administration Building, Room A537, Gaithersburg, MD 20899-0001.

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*tors did not offer their specific weights for each criterion used to judge a program. Finally, evaluators were not asked to provide a judgment of their depth of knowledge of other programs.*

*Since the criteria for determining what constitutes a high-quality entrepreneurship program is, at present, rather fluid and indeterminate, we thought it appropriate to borrow insights from a highly successful and visible evaluation effort in higher education, the education pilot criteria for the Malcolm Baldrige National Quality Award (MBNQA). For an MBNQA evaluation, organizations are assessed across 28 requirements that are embodied in seven categories.*

Leadership. *This category examines senior administrators' commitment and involvement in creating and sustaining performance excellence that has a student focus, clear goals, and high expectations. In the context of entrepreneurship education, the leadership category entails describing the involvement and commitment of entrepreneurship program directors, business school deans, university administrators, advisory board members, and student representatives.*

Information and Analysis. *This category examines how data and information are used to support the overall mission of the program. The focus of this category is towards identifying and specifying information and data that would be appropriate for evaluating the quality of an entrepreneurship program, as well as for making comparisons with other programs. We suggest that entrepreneurship programs begin to systematically collect information about issues such as demographic and performance measures of incoming students enrolling in entrepreneurship courses; comparative information on entrepreneurship, business school and university students; descriptions of the outcomes that specific entrepreneurship courses intend to generate and the measures of the efficacy of each course; and measures of the intended outcomes of the entrepreneurship program in terms of student performance, student satisfaction, and impact on the community (i.e., number of start-ups, students employed in new firms, students working in positions assisting new firms).*

Strategic and Operational Planning. *This category focuses on how a program sets strategic directions and key planning requirements. For an entrepreneurship program, such a requirement would entail generating a strategic plan that specifies the purpose and mission of the program, key student and overall program performance requirements, external factors impacting the implementation of the plan, internal resources and university barriers to change, and key critical success factors.*

Human Resource Development and Management. *This category examines how faculty and staff are supported and developed so as to satisfy the strategic goals of the program. While an entrepreneurship program might typically measure "faculty productivity" as an indicator of this category, the intention is actually towards specifying the resources and systems that impact the ability of staff and faculty to be productive.*

Educational and Business Process Management. *This category specifies key aspects of the design and delivery of the educational research and service components of a program, as well as an examination of the processes involved in improving these components. Rather than programs being compared to each other by the quantity of courses offered, this category requires that programs be measured on the logic, coherency, and efficacy of the educational experience that entrepreneurship students undertake.*

School Performance Results. *This category examines the outcomes of a program, such as student performance and improvement, improvement in services provided by the program, and faculty productivity. This category accounts for 23% of the total evaluation score. The primary focus of this category is determining the improvements in student performance. Such key measures might include student performance in specific courses, student demonstrations of key skills and knowledge through portfolios of original work that*

*they create, measures of student satisfaction, and impact on the community (i.e., number of start-ups, students employed in new firms, students working in positions assisting new firms).*

Student Focus, and Student and Stakeholder Satisfaction. *This category describes the process for determining student and stakeholder needs and expectations, as well as making comparisons of student and stakeholder satisfaction among other programs. This category accounts for 23% of the total evaluation score.*

*The MBNQA evaluation scheme forces us to become aware of the implicit goals, objectives, and pedagogical perspectives of our programs. We must not lose sight of the fact that entrepreneurship programs are and will be evaluated, and that we must, therefore, be ready to offer criteria that we want our programs to be evaluated on. If university entrepreneurship educators do not step forward to assume leadership of our own field, others will surely come to the forefront to determine the rules of the game.* 

## INTRODUCTION

This article discusses some of the logic involved in the development of criteria that might be useful for evaluating university entrepreneurship programs. Our intention is to surface some of the dilemmas and issues inherent in attempting to make comparisons across widely varying pedagogies and curricula. In addition, we hope to redirect the evaluation of entrepreneurship programs away from a dependence on what we perceive to be public relations efforts by many schools, and towards the use of discernible and measurable criteria that can evaluate the important aspects of an entrepreneurship program. We conclude the article with a discussion of criteria that may be useful for measuring progress in entrepreneurship education.

Are there criteria that one might utilize to make comparisons among various universities and colleges offering programs in entrepreneurship? While "Best College" lists are typically based on self-reported information that have been shown to be of questionable validity, and on information that is quantifiable but not necessarily germane to determining the quality of a university education (Gilley 1992), the public has a nearly insatiable demand for any information that might help them make better choices among all of the universities and colleges offering degrees. An interest in college rankings is reasonable and rational. A college degree is an expensive investment for most individuals and families. Prospective students want to have some inkling of the kinds of features (e.g., small class sizes, world-renowned faculty, state-of-the-art facilities, etc.) that might differentiate among a "good, better, or best" education. A ranking might, therefore, be an appropriate measure of some of the criteria that could be used to make a distinction among different university programs. We are now seeing college-level entrepreneurship programs being evaluated with the same kinds of schemes used by magazines to evaluate other university programs.

For example, in the past two years, *Success Magazine* (Callan and Warshaw 1994, 1995) has ranked college-level entrepreneurship programs on evaluation of criteria that seem, on face value, to be reasonable. Colleges and universities are measured on four areas: (1) qualifications of faculty, (2) the variety and depth of the entrepreneurship curriculum, (3) academic standards and student scores, and (4) the quality and depth of resources. Yet, upon a closer look at the specific criteria measured in this survey, such measures as the GMAT (Graduate Management Aptitude Test) scores and GPA (Grade Point Average) scores of incoming students may not be a reasonable measure of these students' entrepreneurial capabilities. A measure that is used to predict success in school (i.e., GMAT) may be an inappropriate measure to use to predict success at starting and growing a business. In addition, measuring the quality and depth of resources at a university by measuring, for example, the number of computers per capita, may be easily quantifiable, but not necessarily correlated to improving the teaching and acquisition of entrepreneurial knowledge and skills. There are many specific measures that are used in these *Success Magazine* rankings that have a tenuous link to factors that might enhance the acquisition of entrepreneurship knowledge, skills, and abilities. A ranking of entrepreneurship programs might therefore be based on information that, while well-meaning, might be a very poor reflection of the relative quality of each specific entrepreneurship program. In the belief that entrepreneurship educators have important insights and a vested interest in the criteria used to evaluate entrepreneurship programs, we think it useful to begin a public discussion of some of the issues involved in generating these rankings.

Concurrent with an advisory role in the *Success Magazine* evaluation of university-level entrepreneurship programs, the first author had undertaken a survey that focused on the development of programs in entrepreneurship at business schools around the world. Over the last 20 years of surveys of entrepreneurship courses (Vesper 1974, 1975, 1976, 1978, 1979, 1980, 1985, 1990, 1993), a growing number of business schools have developed a series of courses in entrepreneurship, variously labeled as either a program, concentration, or major in entrepreneurship. This survey sought to discover what courses might comprise a program in entrepreneurship, as well as other elements that might impact the quality of an entrepreneurship program, such as faculty activities, community outreach, resources applied to entrepreneurship, and perspectives on measures of program efficacy and outcomes.

## A BRIEF HISTORY OF UNIVERSITY-LEVEL ENTREPRENEURSHIP EDUCATION

Until 1970, very few universities offered entrepreneurship courses. The Harvard Business School introduced an entrepreneurship course in 1945, apparently in response to students who were returning from World War II military service to an economy that was in transition due to the collapse of the weapons industries. The course took hold and grew in popularity, although the tenure-track faculty member who began it apparently saw insufficient academic future in it and shifted his attention to the study of boards of directors in major corporations.

The subject of entrepreneurship was not generally fashionable in the decades that followed. This, in part, is reflected in the measures of the entrepreneurial activity in the United States economy during this time. During the 1950s and 1960s, the number of Schedule C's (Income from Business or Profession) filed with the U.S. Internal Revenue Service and the number of corporations per capita experienced a period of steady decline (Gartner and Shane 1995). The big corporations grew bigger.

But by 1970 the number of business schools offering courses in entrepreneurship had begun to change dramatically. Suddenly, there were 16 universities with such courses, of which 12 had started in the preceding two years. What caused this change is difficult to pinpoint. Measures of entrepreneurial activity, such as the number of Schedule C's and the number of corporations per capita stopped falling around 1969 and began to rise. New magazines celebrating entrepreneurship began to emerge: *Inc.*, *Venture*, *In-Business*, and *Entrepreneur*. Connotations of the term "entrepreneur" began to shift from notions of greed, exploitation, selfishness, and disloyalty to creativity, job creation, profitability, innovativeness, and generosity. The venture-capital community, while getting its start just after World War II, became established in the late 1960s as a professional occupational area of high visibility beckoning potential entrepreneurs to try entrepreneurship. Certain industry sectors, such as electronics, had cultivated a population of individuals who could create new products, both commercial and consumer, to put invested capital to work profitably. The advent of the microcomputer accelerated this trend, helping entrepreneurs by making possible the operation of businesses with greatly reduced economies of scale, while at the same time creating opportunities for new software firms that were low in capital intensity and high in margin, therefore, easy to start on a wave of high demand.

From thus base of 16 universities and colleges offering entrepreneurship courses in 1970, the number of schools offering entrepreneurship courses had grown to over 400 by 1995. As the number of schools offering entrepreneurship courses grew, so did the number of schools offering more than one course in entrepreneurship. There began to be programs in this subject.

## WHAT IS AN ENTREPRENEURSHIP PROGRAM?

There appear to be at least 50 universities that offer four or more courses clustered in the area of entrepreneurship that allow students to take concentrations, majors, or degrees (Gartner and Vesper 1994). We recognize that there is a diversity of views among academics about what constitutes "entrepreneurship" as a field of study (Gartner 1990), as well as in what constitutes an entrepreneurship program. For example, entrepreneurship scholars have differing views on whether entrepreneurship must focus on organization creation, growing firms, innovation, value creation, and ownership. While previous surveys by Vesper (1974–1993) have defined entrepreneurship as business entry, whether by start-up or acquisition and whether independently or within an established organization, the growth of university-level entrepreneurship programs has tended to broaden this viewpoint to include other topics such as family business, managing smaller enterprises, and managing high-growth businesses. As a way to categorize the variation in types of university-level efforts in entrepreneurship education, Plaschka and Welsch (1990) have suggested four different dimensions: number of courses offered (single to multiple), degree of integration (low to high), stages of business transition (e.g., inception, survival, growth, expansion, maturity), and number of disciplines. Their article suggested that entrepreneurship education is moving towards integrative, comprehensive, and holistic programs, that is, towards the vision of business education offered in Porter and McKibbin (1988).

It is not difficult to see how an initial focus on entrepreneurship as "a person who organizes and manages a business undertaking, assuming the risk for the sake of profit" (Webster), could expand into other topic areas. The process of starting a business introduces the possibility of starting a business with zero profit (non-profit entrepreneurship) or that of starting a new business within an existing business (corporate entrepreneurship). Linked to the process of start-up is the aftermath of start-up: small and growing businesses. The association of entrepreneurship with business entry also suggested to some schools a link between entrepreneurship and family business. Starting a business has family and estate implications. Running a family business might also have small-business implications. Inheriting a business could be connected to the issues of business entry.

The popularity of the term "entrepreneurship" has also propelled its application as an adjective. The term "entrepreneurial" has found a widening application, connoting innovativeness, initiative, job creation, creativity, ambition, perseverance, achievement, and success (Lumpkin and Dess 1996).

Finally, many wealthy individuals have given money to universities to promote entrepreneurship, and therefore many universities have sought to seek more funding using the "entrepreneurship" label. Sometimes entrepreneurship programs are created with this money, sometimes not, and in between are many combinations.

Given the variety of meanings of "entrepreneurship," the variety of different courses in entrepreneurship, and the variety of ways that the entrepreneurship label is used by universities to promote differing agendas, finding a reasonable way to rate and rank entrepreneurship programs is a challenge.

## RATING ENTREPRENEURSHIP PROGRAMS

Scholars debate about whether entrepreneurship programs should be rated and ranked: Every program is unique, and to some degree, incomparable. No two people would rate programs in exactly the same way. Aggregate results of ratings apply to no single program in particular. Programs are constantly changing, so ratings are constantly out of date. Criteria for ratings also change and are also out of date. Some schools are more visible than others, and nobody knows them all. Ratings are bound to be distorted by ignorance. Even so, rankings of entrepreneurship programs have occurred and seem to occur with greater frequency.

A number of magazines such as *Success* and *Business Week* have started rating and ranking entrepreneurship programs. (See Table 1.) These magazines will continue to do so because their readership wants this information and will buy these magazines to get it. Publication of ratings and rankings have powerful effects on universities. High-ranked programs receive more inquires, applications, and enrollments. Schools that depend on tuition income pay close attention to applications and enrollments, and therefore pay attention to how they fare in these ratings. Hence, ratings influence what schools offer.

Ratings and rankings can be constructive. Programs that make contributions to entrepreneurship education deserve to be recognized for it. If some programs have better features, then recognizing these programs through widely distributed publications can let other schools learn these better ways of operating. Competitive scrutiny can help ward off complacency and stimulate universities to strive for better ways of operating. Finally, public scrutiny and exposure can help weed out misleading advertising among the claims of different programs. Since ratings and ranking are bound to influence the direction of entrepreneurship programs, it makes sense to consider, with care, how the ratings should best be done to enhance the benefit of their influence.

A mail survey was undertaken in late 1994 to learn about entrepreneurship programs around the world. This survey sought to discover how universities ranked other university entrepreneurship programs, how universities determined what courses constituted a program in entrepreneurship, and how universities formed opinions about other criteria that might impact the quality of an entrepreneurship program, such as faculty activities, community outreach, resources applied to entrepreneurship, and perspectives on measures of program efficacy and outcomes. The survey was sent to 941 business schools in the United States, 42 in Canada, and 270 overseas. We recognize that this survey is not a complete and comprehensive survey of all four-year colleges and universities that might have an entrepreneurship program. But it is our opinion that the survey represents a broad and thorough exploration of entrepreneurship courses and programs in existence.

Although the questionnaires were addressed to deans, they were answered by a variety of people, sometimes deans, and sometimes others to whom a dean had passed the question-

**TABLE 1** Top-Rated U.S. Entrepreneurship Programs

| Rank | School | Academic Points | Entrepreneur Magazine 1993 | Business Week 1993 | Success 1994 Top 25 | Success 1995 Top 25 | Success 1995 2nd 25 |
|---|---|---|---|---|---|---|---|
| 1 | Babson College | 1063 | X | X | X | X | |
| 2 | Harvard Business School | 904 | X | X | X | X | |
| 3 | Wharton School | 810 | X | X | X | X | |
| 4 | Univ. of Southern California | 499 | X | X | X | X | |
| 5 | Univ. of Texas, Austin | 421 | | X | X | X | |
| 6 | U.C.L.A. | 408 | | X | X | X | |
| 7 | Wichita State | 362 | X | | X | | X |
| 8 | Univ. of Georgia | 298 | | X | X | X | |
| 9 | Carnegie-Mellon | 290 | | X | X | X | |
| 10 | Northwestern | 283 | X | X | X | | X |
| 11 | New York Univ. | 281 | | X | X | X | |
| 12 | Rensselaer | 262 | X | X | X | X | |
| 13 | Univ. of St. Thomas | 245 | | | X | X | |
| 14 | Baylor Univ. | 224 | X | | | X | |
| 15 | DePaul Univ. | 215 | | X | X | X | |
| 16 | Stanford | 176 | | X | | | X |
| 17 | Kennesaw State Univ. | 162 | | X | | | X |
| 18 | Ball State Univ. | 159 | X | X | | X | |
| 19 | Univ. of Arizona | 147 | X | X | X | | X |
| 20 | Case-Western Univ. | 136 | | | X | | X |
| 21 | Cornell Univ. | 133 | | | X | X | |
| 22 | Univ. of Minnesota | 116 | | | X | X | |
| 23 | Univ. of Colorado, Boulder | 101 | | | X | X | |
| 24 | San Diego State Univ. | 97 | | X | | | |
| 25 | Boston Univ. | 79 | | | X | X | |
| 26 | U. C., Berkeley | 72 | | | X | X | |
| 27 | Univ. of So. Carolina | 66 | | | X | X | |
| 28 | Univ. of Washington | 62 | | | | | X |
| 29 | Fairleigh Dickinson Univ. | 61 | | | X | | X |
| 30 | St. Louis Univ. | 59 | | | | | X |
| 31 | Southern Methodist Univ. | 58 | | X | | | |
| 32 | Univ. of Maryland | 38 | X | | X | X | |
| 33 | Purdue Univ. | 38 | | | | | |
| 34 | Illinois, Chicago | 35 | | | | | X |
| 35 | Georgia Tech | 26 | | | | | |
| 36 | Univ. of Illinois, Urbana | 20 | | X | | | |
| 37 | Thunderbird | 20 | | | | | X |
| 38 | Fresno State | 18 | | | | | |
| 39 | Seattle Pacific Univ. | 18 | | | | | |
| 40 | Baldwin-Wallace | 17 | | | | | |
| 41 | Duke Univ. | 16 | | | | | |
| 42 | Univ. of Oregon | 13 | | | | | X |
| 43 | Miami, Ohio | 13 | | | | | |
| 44 | Ohio State | 13 | | | | | |
| 45 | Columbia Univ. | 9 | | | | | X |
| 46 | Univ. of Montana | 9 | | | | | |
| 47 | Univ. of So. Dakota | 8 | | | | | |
| 48 | M.I.T. | 7 | | | | | X |
| 49 | Rice Univ. | 3 | | | | | |

(*continued*)

**TABLE 1** (Continued)

| Rank | School | Academic Points | Entrepreneur Magazine 1993 | Business Week 1993 | Success 1994 Top 25 | Success 1995 Top 25 | Success 1995 2nd 25 |
|---|---|---|---|---|---|---|---|
| 50 | Univ. of Nebraska | 2 | | | | | X |
| 51 | Central Arkansas | 1 | | | | | |
| | Brigham Young | 0 | | | | X | |
| | Birmingham Southern | 0 | | | | | X |
| | Cal Poly Pomona | 0 | | | | | X |
| | Cal State Hayward | 0 | | | | | X |
| | Canisius College | 0 | | | | | X |
| | George Washington Univ. | 0 | | | X | X | |
| | Georgetown Univ. | 0 | | | | X | |
| | Indiana, Bloomington | 0 | | | | | X |
| | Iowa | 0 | | | | | X |
| | James Madison | 0 | | | | | X |
| | Samford Univ. | 0 | | | | | X |
| | San Francisco State | 0 | | | | | X |
| | Wisconsin, Whitewater | 0 | | | | X | |
| | Xavier | 0 | | | | | X |

naire. Of the 311 replies, 233 came from U.S. business schools, 16 from Canadian schools, and 62 from schools in other countries.

## RANKING ENTREPRENEURSHIP PROGRAMS

Respondents were given a table that listed in three columns the schools whose entrepreneurship offerings have been noted as most prominent by three magazines: *Success* (1994), *Business Week* (1993), and *Entrepreneur* (1993). The schools were listed alphabetically. A fourth column of the table was provided so that respondents could rank the schools they regarded as the top ten besides their own. Space was also provided for "write-ins." Some respondents ranked 10 schools. Some simply checked 10 schools without listing a rank. Other respondents included more than 10 schools. Of the 311 returned questionnaires, 62 provided ranking information. In all, respondents either ranked, checked, and/or nominated a total of 51 U.S., plus 4 Canadian, and 9 overseas schools.

To combine the different types of responses consistently, a weighting scheme was used under which a school ranked #1 by a respondent was assigned 30 points, #2 was assigned 29 points, #3 received 28 points, etc. If a school was only checked by a respondent, it was given the same number of points as if it has been the middle-ranked school on the array of schools that the respondent gave checks to. Self-rankings did not count. How schools ranked in total points allocated by this scheme can be seen in Tables 1, 2, and 3.

Table 1 presents the rankings of U.S. schools, Table 2 presents the rankings of Canadian schools, and Table 3 the rankings of schools of other countries. The reason for this separation is that the majority of respondents were from U.S. schools, and the comments that some of the respondents made, such as "I haven't a clue," indicated that their familiarity with U.S. schools was limited and their knowledge of schools in other countries was still less. The magazines whose published ratings were included in the questionnaire were all published

**TABLE2** Top-Rated Canadian Entrepreneurship Programs

| Rank | School | Academic Points | Entrepreneur Magazine 1993 | Business Week 1993 | Success 1994 Top 25 | Success 1995 Top 25 | Success 1995 2nd 25 |
|---|---|---|---|---|---|---|---|
| 1 | Univ. of Calgary | 55 | X | | | | |
| 2 | HEC, Montreal | 16 | | | | | |
| 3 | York Univ., Toronto | 9 | X | | | | |
| 4 | Univ. of Western Ontario | 8 | | | | | |

in the United States, and all but one of them listed only U.S. schools. So there was clearly both a national bias and also imperfect knowledge of what was going on at other schools in this field.

The schools appearing last on the list in Table 1 all failed to receive any nominations to the "top ten" by respondents, yet they appear in the listing produced subsequently by *Success Magazine* in 1995. This is perhaps due, in part, to the fact that the two pollings took place at different times and both the offerings of schools and the knowledge that academics at other schools have about other programs are changing fairly fast. This flux in rankings seems likely to continue.

A number of issues should be noted in regard to these academic rankings. First, since the ranking form included rankings from three previous magazine evaluations, it is likely that these previous benchmarks may have influenced the academic responses. Second, as noted earlier, we received many written comments from academics indicating the paucity of information they had available for use in ranking various entrepreneurship programs. Except for the compendia of entrepreneurship course descriptions published by Vesper (1974–1993) and the Solomon (1986) national survey, little systematic evidence that compares and describes entrepreneurship courses and programs has been published and disseminated. It would seem likely that many of the academics in this survey are not familiar with this information, and would only have knowledge of other programs through anecdotal evidence or through newsletters, brochures, and flyers distributed by some of these programs. To the extent that the knowledge level of many of the academics who ranked the various programs is incomplete, we think it appropriate to view these rankings with some reservation. Finally, this survey did not provide direct links between the rankings and facts relevant to subsequent criteria thought to be important for ranking programs. For example, while

**TABLE 3** Top-Rated Entrepreneurship Programs Abroad

| Rank | School | Academic Points | Entrepreneur Magazine 1993 | Business Week 1993 | Success 1994 Top 25 | Success 1995 Top 25 | Success 1995 2nd 25 |
|---|---|---|---|---|---|---|---|
| 1 | Durham Univ., UK | 29 | | | | | |
| 2 | Swinburne, Australia | 28 | | | | | |
| 3 | Univ. of Stirling, UK | 27 | | | | | |
| 4 | Vaxjo, Sweden | 17 | | | | | |
| 5 | INSEAD, France | 17 | | | | | |
| 6 | Vincennes Univ., France | 16 | | | | | |
| 7 | Cranfield Institute, UK | 16 | | | | | |
| 8 | London Business School, UK | 14 | | | | | |
| 9 | Bocconi, Italy | 1 | | | | | |

**TABLE 4** Ranking of Program Criteria by Other Academics

| Criterion | Overall rank $n$=130 | U.S. rank $n$=112 | Non-U.S. rank $n$=28 |
|---|---|---|---|
| 1. Courses offered | 1 | 1 | 1 |
| 2. Faculty publications | 2 | 2 | 2 |
| 3. Impact on community | 3 | 3 | 3 |
| 4. Exploits of alumni | 4 | 4 | 4 |
| 5. Innovations | 5 | 5 | 5 |
| 6. Alumni start-ups | 6 | 6 | 6 |
| 7. Outreach to scholars | 7 | 7 | 7 |
| 8. Competitions and awards won | 8 | 8 | 13 |
| 9. Years of activity | 9 | 9 | 12 |
| 10. Size of MBA program | 10 | 10 | 11 |
| 11. Halo of school or university | 11 | 14 | 8 |
| 12. Magnitude of resources | 12 | 11 | 15 |
| 13. Alumni comments years later | 13 | 13 | 14 |
| 14. Size of undergrad program | 14 | 16 | 12 |
| 15. Incoming student qualities | 15 | 15 | 9 |
| 16. Size of doctoral program | 16 | 16 | 10 |
| 17. Faculty start-ups | 17 | 16 | 17 |
| 18. Location | 18 | 18 | 18 |

academics indicated that "courses offered" was the most important criterion for ranking an entrepreneurship program, limitations of time and resources have not yet allowed investigation of the correlation between ranks given and actual numbers or the makeup of courses offered by each program.

## RANKING CRITERIA

In order to determine how academics made judgments about the relative quality of various entrepreneurship programs, respondents were asked to indicate, with a rank ordering, what criteria should be used for rating entrepreneurship programs. The results can be seen in Table 4. The leading criteria are courses offered, faculty publications, impact on community, and innovations and accomplishments of alumni. The figures in Table 4 were computed by weighting responses in order of rank, with rank one receiving the highest weighting, and then adding the weights. The top rank was given 25 points, the next 24, and so forth, for as many criteria as the respondent ranked.

How quickly the rankings dropped off from higher ranked to lower ranked criteria can be seen from the histogram (Figure 1), which plots the total points for each of the criteria. From this it can be seen that courses offered was strongly the number-one criterion, with the next few substantially below it but fairly closely ranked to each other.

### Courses Offered

The number-one criterion at the time of this survey, "courses offered," may seem enigmatic if described in those two words alone. Does it mean number of different courses, size of classes, number of credits or class sessions, something about the way the courses are taught, who is doing the teaching, or some other feature of the courses? Respondents were asked which of 22 types of courses (plus a blank for other write-ins) the respondent's school either

**TABLE 5** Number of Schools with Different Courses in Catalog or Planned

| | Undergraduate | | Graduate | |
|---|---|---|---|---|
| Courses offered or planned | In catalog | Planned | In catalog | Planned |
| Entrepreneurship or starting new firms | 128 | 14 | 98 | 5 |
| Small business management | 109 | 5 | 31 | 2 |
| Field projects/venture consulting | 44 | 4 | 31 | 3 |
| Starting and running a firm | 34 | 2 | 18 | 1 |
| Venture plan writing | 31 | 3 | 23 | 1 |
| Venture finance | 30 | 4 | 30 | 6 |
| Entrepreneurship for non-business majors | 18 | 5 | 7 | 1 |
| Family business | 16 | 6 | 12 | 3 |
| Venture opportunity finding/screening | 15 | 1 | 9 | 0 |
| Venture marketing | 13 | 0 | 12 | 5 |
| Management of fast-growing firms | 11 | 4 | 10 | 4 |
| Venturing in (arts, nursing, Eastern Europe, technology, or other fields) | 11 | 3 | 6 | 1 |
| Creative thinking | 10 | 1 | 7 | 2 |
| Franchise development | 9 | 3 | 9 | 2 |
| International venturing | 9 | 2 | 6 | 0 |
| Law for entrepreneurs | 9 | 5 | 5 | 3 |
| Innovation evaluation | 7 | 0 | 21 | 1 |
| Technology transfer | 6 | 2 | 22 | 4 |
| Entrepreneurship for (bankers, software writers, biologists, or other fields) | 5 | 1 | 4 | 2 |
| Corporate venturing | 4 | 1 | 15 | 6 |
| Business entry via acquisition | 3 | 2 | 8 | 1 |
| Street smarts in business | 2 | 0 | 2 | 0 |
| Other (please attach catalog description) | 20 | 0 | 18 | 1 |

had (as indicated by an established catalog number) or was planning to add. This list appears in Table 5. A challenge in interpreting the results for tabulation arose from the fact that while some respondents listed the catalog number, others simply put check marks. Our decision in the latter event was to regard the data as indicating that there was one course that treated all the topics checked. This may have resulted in under-counting the actual number of courses that were offered.

If the same catalog number was used for more than one of the courses on the list, then it was considered a single course for tabulating purposes and credit for that course was divided evenly among however many courses that number was used for. Sometimes a catalog number was the same for both undergraduate and graduate courses except for the first digit. This might mean there were two courses or it might mean that one course was offered to both categories of students, with undergraduates signing up under one number and graduate students under a different one for the same series of classes. Not being able to tell whether this is the case, we counted such courses as separate. This may have resulted in over-counting the actual number of different graduate and undergraduate courses offered.

Aside from the above difficulties of determining how many different courses were offered, there is another problem associated with using the number of different courses offered as a criterion for rating programs. It implies that entrepreneurship as a subject should be taught in separate courses, rather than as part of "regular" courses or infused throughout the curriculum. Not all schools see it that way. Some claim that the subject is covered as part of other courses. Should there be a second rating scheme for schools that take such an approach? But then, how should schools be evaluated that sometimes treat the subject with separate courses and other times infuse it in regular courses? Should that make a third category, or possibly a whole array?

## Faculty Publications

The importance of the second criterion, faculty publications, was significantly lower that that of courses offered. Unfortunately, there was no question that asked which kinds of publications (such as books and journal proceedings, and magazine articles) or what about the publications (such as how often cited, whether data-based, how applied, nature of target audience, length) should be considered of most importance.

## Impact on Community

The impact that the entrepreneurship program might have on the community ranked closely behind faculty publications in importance. Public symposia, student consulting projects, and company spin-offs from the university would be examples of this criterion. To determine which of these was most effective or valuable in evaluating an entrepreneurship program will require further investigation. Judging from the indicated importance of this criterion, however, such a study might be worthwhile. Foundations interested in the practical accomplishments of entrepreneurship programs might consider underwriting such a study.

## Alumni Exploits

Alumni exploits include not only start-ups by alumni, but also alumni participation in the ventures of others as investors, partners, employees, or other helpers. Determining the number of graduates who have started companies has, to date, been beyond the resources of most schools. The few informal studies done by schools show a strong correlation between students who study entrepreneurship and, at some later time, start companies. It will never be possible to separate selection from training as the cause of these start-ups, but it should be possible, through detailed case studies, to uncover which elements of the educational process influence selected decisions and actions in alumni start-ups.

## Innovations

This criterion could be interpreted either to mean innovations of the alumni of these programs or innovations in the programs, themselves. This criterion was ranked virtually on a par with alumni exploits. The conclusion for schools might be simply to regard introduction of innovations as important, whether in their own activities or those of their students. Part of the self-critique of a school desiring to be high-rated would be to ask, "What innovations have we been responsible for?" Innovative programs would seem more likely to be noticed.

## Alumni Start-ups

Although not the same thing as "alumni exploits," start-up accomplishments could, logically, be included within it.

## Outreach to Scholars

This criterion seems to refer to such program activities as hosting conferences, sponsoring journals, creating and distributing new teaching materials, publishing newsletters, and helping other schools. Faculty publications could be part of this.

## Other Criteria

The following were all emphasized about equally as still important, but noticeably less so than the first seven criteria: awards and national competitions won, years in the business, number of MBA students, halo of the school, magnitude of resources, comments of alumni years later, number of undergraduate students, and qualities of incoming students. It is clear that performance on some of the "other criteria" is likely to influence performance on the first seven criteria. For instance, the magnitude of resources can affect a program's ability to score on any of the other dimensions. More years "in the business" (the length of time the program has existed) will have allowed time for more accomplishments to have occurred. A larger number of students, whether MBA or undergraduate, will provide a bigger population for luck, if nothing else, to work in producing alumni exploits.

## Least Important Criteria (for now?)

Finally, the criteria judged least important by respondents in rating programs were the following: number of doctoral students, faculty start-ups, and location. Low importance ratings on any of these criteria could be regarded with some surprise. The late provost emeritus of Stanford University, Frederick Terman, regarded Ph.D. production as the prime criterion for elevating the status of his engineering school, having noted from his earlier association with Harvard and MIT, that Ph.D. production had been a winning strategy for those schools. The rating of faculty start-ups as a relatively unimportant criterion for judging entrepreneurship programs was also a surprise to us. It would seem that the creation of a new business would be a very relevant demonstration that an entrepreneurship faculty member had some practical knowledge to teach from. Finally, it would seem that location should be important for judging a program. It has long been observed that exceptionally high-performance entrepreneurship tends to cluster geographically. That is how "Route 128"and "Silicon Valley" became familiar terms. Presence of role models to serve as speakers and mentors to entrepreneurship students are elements that faculty commonly regard as very helpful in entrepreneurship courses, and such resources are certainly more available in metropolitan areas where, coincidentally, entrepreneurship programs are generally situated.

This study is a first cut at exploring criteria for evaluating entrepreneurship programs. In the future it seems likely that better information and further thought will change both the criteria and their weightings. At the same time, some schools will introduce new programs. Existing programs will expand, refine, and publicize their offerings. As all this happens, and people learn about these changes and innovations, the rankings of entrepreneurship programs should change as well, and become more meaningful. In the meantime, in the spirit of generating some discussion on this issue, we offer a framework for evaluating entrepreneurship programs and some opinions on specific criteria for measuring progress in entrepreneurship education.

## DISCUSSION

As we have suggested earlier in this article, the rankings of university entrepreneurship programs should be viewed with much skepticism. There are a number of problems with these rankings. Evaluators did not specify the criteria they actually used to rank entrepreneurship programs, just ones they considered important. Evaluators did not offer their specific weights for each criterion used to judge a program, they just ranked them. Finally, evaluators were not asked to provide a judgment about their depth of knowledge of other programs. Since the criteria for determining what constitutes a high-quality entrepreneurship program is, at present, rather fluid and indeterminate, we suggest that a discussion of measures of quality in entrepreneurship education should be placed in the context of the larger movement in higher education toward achieving quality and continuous improvement (Boyer 1990; Malcolm Baldrige National Quality Award 1995; Seymour 1992) and, particularly, the quality movement in business schools (Meisel and Seltzer 1995; Vance 1993). Rather than inventing evaluation criteria and processes for evaluating entrepreneurship programs, we thought it more appropriate to borrow insights from highly successful and visible evaluation efforts in higher education.

In our review of these evaluation schemes, we believe the Malcolm Baldrige National Quality Award (MBNQA) *Education Pilot Criteria: 1995* offers a very comprehensive and detailed format that may serve as a guide for identifying criteria for measuring quality in entrepreneurship education. The MBNQA was established in 1987 to promote quality improvement, recognize organizations that have made substantial improvements in quality, and foster the sharing of best-practices information. The MBNQA is one of the largest quality-improvement programs. Over one million copies of the MBNQA Criteria have been disseminated, though there have been only 546 applicants for the Baldrige Award and less than 30 award recipients. The MBNQA Education Pilot Criteria are structured in a format similar to all MBNQA programs. Organizations are assessed across 28 requirements that are embodied in seven categories. For the education pilot criteria framework, these seven categories and their weights in the overall evaluation scheme are (1) Leadership, 9%; (2) Information and Analysis, 7.5%; (3) Strategic and Operational Planning, 7.5%; (4) Human Resource Development and Management, 15%; (5) Educational and Business Process Management, 15%; (6) School Performance Results, 23%; and (7) Student Focus and Student and Stakeholder Satisfaction, 23%. The MBNQA is oriented towards helping organizations undertake substantial efforts at self-evaluation and study leading to significant improvements in organizational processes, structure, and outcomes. The MBNQA is primarily a diagnostic system that focuses on achieving results. While the MBNQA evaluation process is very elaborate and detailed, and cannot be easily summarized in a few sentences or pages, it is in the context of the MBNQA format that we offer some suggestions for how entrepreneurship programs might be rated. Further information about the MBNQA Education Pilot Program is provided at the end of this article.

### Leadership

This category examines senior administrators' commitment and involvement in creating and sustaining performance excellence that has a student focus, clear goals, and high expectations. In the context of entrepreneurship education, the leadership category entails describing the involvement and commitment of entrepreneurship program directors, business school deans, university administrators, advisory board members, and student representatives. One way that senior-administrator commitment and involvement could be determined is by evaluating the centrality of entrepreneurship education to the university's mission. For instance, a high-quality entrepreneurship program might have "entrepreneurship" as a significant part of the mission statement of its business school. Entrepreneurship should, indeed, be part of the mission of the university as a whole. Ways that a leadership commitment to entrepreneurship might be measured might include how often entrepreneurship is described in university publications, such as its bulletin, catalogue, recruiting brochures, and other materials, as well as how often entrepreneurship is publicly championed in speeches, presentations, and articles by the key decision-makers in the university. Finally, there should be some emphasis on where an entrepreneurship program is placed in the administrative structure of the university. In many universities, the primary way a topic area gains prominence is through department status. Universities form departments as a way to focus faculty efforts, coordinate resources through a specified budget, and recruit students. Our primary concern in the leadership category is that the quality criteria demonstrate that entrepreneurship is an important focus of the university, and not merely an afterthought or tangential response to placate growing student interest in this topic, or a gesture to garner financial support from successful entrepreneurial alumni to use for other projects of non-entrepreneurship faculty. As evidence from the corporate-venturing literature suggests (Block and MacMillan 1993), a program within a large university is unlikely to flourish and innovate without the active support of top administrators, as well as the championing efforts of business school deans, and the cooperation of the school's voting faculty. The day of the "one-man band" is probably over for significant programs.

## Information and Analysis

This category examines how data and information are used to support the overall mission of the program. The focus of this category is towards identifying and specifying information and data that would be appropriate for evaluating the quality of an entrepreneurship program, as well as for making comparisons with other programs. This includes collection of information for comparisons and benchmarking to improve program performance. As we have indicated throughout this article, one of the major problems in conducting surveys of entrepreneurship programs has been the paucity of information that each program collects. We suggest that entrepreneurship programs begin to systematically collect information about such issues as demographic and performance measures of incoming students enrolling in entrepreneurship courses; comparative information on entrepreneurship, business school, and university students; descriptions of the outcomes that specific entrepreneurship courses intend to generate and measures of the efficacy of each course; and measures of the intended outcomes of the entrepreneurship program in terms of student performance, student satisfaction, and impact on the community (i.e., number of start-ups, students employed in new firms, students working in positions assisting new firms). While much of this information might seen onerous to collect, we posit that the growing trend towards quality management in higher education will require that this information, or information similar to it, be collected from all university units in the foreseeable future. Without such information on inputs, processes, and outcomes, it is difficult to adequately evaluate the performance of any program. The collection of relevant information will be critical for making sound comparisons among various entrepreneurship programs. There should also be a way to share it efficiently to minimize redundant inquiries.

## Strategic and Operational Planning

This category focuses on how a program sets its strategic direction and develops its key planning requirements. The primary emphasis of this category is on aligning the strategic planning process with student needs and expectations, external factors, requirements and opportunities, internal capabilities, and resources, so that improvements can be made in student performance and in the service and research components of the program if they, too, are present. For an entrepreneurship program, such a requirement would entail generating a strategic plan that specifies the purpose and mission of the program, key student and overall program performance requirements, external factors impacting the implementation of the plan, internal resources and university barriers to change, and key critical success factors. The goal of this category is to generate a road map to guide the success of the program. A measure of this category would be whether a program has a strategic plan, or not.

## Human Resource Development and Management

This category examines how faculty and staff are supported and developed so as to satisfy the strategic goals of the program. While an entrepreneurship program might typically measure "faculty productivity" as an indicator of this category, the intention is actually towards specifying the resources and systems that impact the ability of staff and faculty to be productive. What might be measured in this category could be the amount of resources devoted to enabling faculty to do research, helping faculty become better entrepreneurship teachers (through such activities as case writing and attending workshops, conferences, and training programs), and enabling staff to better serve students or work with entrepreneurial clients. The ability of faculty to generate academic publications or high teacher ratings, for example, are only two possible measures of the support provided to conduct such activities. In the MBNQA framework, faculty productivity is the result of efforts to enhance the well-being of faculty. Increasing productivity is difficult without the necessary resources to achieve such gains.

## Educational and Business Process Management

This category specifies key aspects of the design and delivery of the educational research and service components of a program, as well as an examination of the processes involved in improving these components. This is the category in the evaluation scheme where such measures as "courses offered" have some bearing. This category recognizes a more comprehensive and detailed description of a program's pedagogy and curriculum. A high-quality program would articulate the logic of the course offerings (e.g., What is their purpose? What do they hope to achieve? Who are these courses designed for? How do these courses fit with the rest of the business school's curriculum?) and how new courses are designed and introduced. A high-quality program would need to describe how these courses are delivered and how the program measures the success of the delivery of these courses (e..g, student evaluations of courses and instructors, enrollment demand, complaints, attendance rates). A high-quality program would specify the kinds of support services available to help entrepreneurship students succeed (e.g., counseling, advising, internships, mentors). Finally, a high-quality program would show how research, scholarship, and the service activities of faculty are linked to achieving the educational mission of the program. Such a comprehensive description of the pedagogy and curriculum of an entrepreneurship program is a quantum change in the level of detail and thought necessary compared to a measure such as "courses offered." Rather than programs being compared to each other by the quantity of courses offered, this category requires that programs be measured on the logic, coherency, and efficacy of the educational experience that entrepreneurship students undertake.

### School Performance Results

This category examines the outcomes of a program, such as student performance and improvement, improvement in services provided by the program, and faculty productivity. As may have been noticed earlier, this category accounts for 23% of the total evaluation score. The primary focus of this category is determining the improvements in student performance. In order to specify such improvements, a program must identify key measures and indicators of performance. As mentioned earlier, such key measures might include student performance in specific courses, student demonstrations of key skills and knowledge through portfolios of original work that they create, measures of student satisfaction, and impact on the community (i.e., number of start-ups, students employed in new firms, students working in positions assisting new firms). For research universities, programs should also be evaluated on faculty contributions to knowledge creation and transfer. Measuring the number of faculty publications is one, rather limited, approach to describing faculty productivity. Text books, professional and trade books, cases, teaching exercises, and activities such as consulting, counseling, and advising students and entrepreneurs might also be appropriate measures for certain university programs. Finally, the impact of a program's contribution to knowledge should not just be measured by counting the number of times a journal article is cited in a citation index, but also through evidence of impact on practitioner thought and action. Since entrepreneurship is an occupation, impact should ultimately be measured as an effect on practitioner activities.

### Student Focus, and Student and Stakeholder Satisfaction

This category describes the process for determining student an stakeholder needs and expectations, as well as making comparisons of student and stakeholder satisfaction among other programs. This category accounts for 23% of the total evaluation score. A high-quality entrepreneurship program would seek to determine the needs and expectations of current students, as well as future student needs and expectations, and the needs and expectations of other key stakeholders (e.g., parents, communities, alumni, employers, legislatures, and practicing entrepreneurs). In the MBNQA evaluation process, the concept of quality is determined by a thorough understanding of student and stakeholder needs. In many respects, the MBNQA process begins with this category. The quality of a program, is therefore, externally driven, rather than driven by what we, as educators and administrators, might want to posit as a key and important contribution. An entrepreneurship program should be "close to the customer," and information should be collected that demonstrates what students and stakeholders need and expect, as well as how a program best attempts to serve them.

## CONCLUSIONS

The MBNQA is one approach to a comprehensive and detailed evaluation of an entrepreneurship program. What the MBNQA evaluation scheme does is force us to become aware of the implicit goals, objectives, and pedagogical perspectives of our programs. Our measures of the quality of entrepreneurship programs have to become more comprehensive and robust to reflect these issues. We do not know of any entrepreneurship program that has undertaken the kind of evaluation specified in the MBNQA process. But we think such efforts should help the refinement process. This may be the "higher road" towards improving entrepreneurship programs, rather than seeking to play the "ratings game" of satisfying superficial criteria. The evolution of entrepreneurship programs in colleges and universities is still in its infancy. More debate and dialogue among academics, administrators, students, and other stakeholders must be encouraged. The criteria for determining high-quality entrepreneurship education programs should not become fixed at this time. Yet, we must not lose sight of the fact that entrepreneurship programs are and will be evaluated, and that we must, therefore, be ready to offer criteria that we want our programs evaluated on. If university entrepreneurship educators do not step forward to assume leadership of our own field, others will surely come to the forefront to determine the rules of the game.

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